

Statistics Lab 52568 — Formula & Concept Sheet

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Sequence Basics (Lab 1)

Chargaff: $A \approx T$, $C \approx G$. **GC content:** $gc_k = (G_k + C_k)/b \in [0, 1]$.

Coverage & the Poisson Model (Lec 2–3)

Y_i = number of reads starting at base i , for $i = 1, \dots, I$.

J = total number of fragments in the experiment.

Average coverage:

$$\lambda = \frac{J}{I} = \frac{\text{total reads}}{\text{total bases}}, \quad \text{avg distance} = \frac{1}{\lambda}$$

Uniform Poisson — 3 assumptions:

1. Identical distribution: every fragment has the same p_i
 2. Uniformity: $p_i = 1/I$ for all i
 3. Independence: fragments land independently
- (1)+(3): $Y_i \sim \text{Binomial}(J, p_i)$. Adding (2) + Poisson approx (J large, $1/I$ small):

$$Y_i \sim \text{Poisson}(\lambda), \quad \mathbb{E}[Y] = \text{Var}[Y] = \lambda$$

Additive property: $X_1, \dots, X_K \sim \text{Pois}(\lambda)$ independent $\Rightarrow X_1 + \dots + X_K \sim \text{Pois}(K\lambda)$.

$\Rightarrow$ Bin of size b : $Y_k \sim \text{Pois}(\lambda_k)$, $\lambda_k = \sum_{i \in k} p_i \cdot J$.

VMR (Variance-to-Mean Ratio):

$$\text{VMR} = \frac{\text{Var}[Y]}{\mathbb{E}[Y]} = 1 \quad \text{under uniform Poisson}$$

Overdispersion & Variance Decomposition (Lec 3)

Real data: $\text{VMR} \approx 14.7$ — uniform model fails. Why?

Two-level model: $Y_k \sim \text{Pois}(\lambda_k)$, each bin has its own rate.

Law of Total Variance (exact identity):

$$\text{Var}(Y) = \underbrace{\mathbb{E}[\text{Var}(Y | \lambda)]}_{\text{avg within-group}} + \underbrace{\text{Var}(\mathbb{E}[Y | \lambda])}_{\text{variance of group means}}$$

Applied here:

$$\text{Var}(Y_k) = \underbrace{\mathbb{E}[\lambda_k]}_{\bar{\lambda} \text{ (Poisson noise)}} + \underbrace{\text{Var}(\lambda_k)}_{\text{rate heterogeneity}}$$

$$\text{VMR} = 1 + \frac{\text{Var}(\lambda_k)}{\bar{\lambda}}$$

$\text{VMR} = 1$ iff all λ_k equal. Spatial correlation of GC prevents $\text{Var}(\lambda_k)$ from averaging out under binning — rates in nearby bins are similar.

Robust estimators: Center: median. Spread: IQR, $\text{MAD} = \text{med}|x_i - \text{med}(x)|$. Use instead of mean/SD when data are skewed or have outliers.

B&S (2012): GC bias unimodal (peaks $\sim 50\%$); driven by full-fragment GC (PCR cause); MAD 9.5 vs loess 10.8; fit per sample.

From histogram (midpoints m_j , counts n_j , $n = \sum n_j$):

$$\hat{\mathbb{E}}[Y] = \frac{\sum m_j n_j}{n}, \quad \widehat{\text{Var}}(Y) = \frac{\sum m_j^2 n_j}{n} - \hat{\mathbb{E}}[Y]^2$$

Variance decomp plug-in: $\hat{\mathbb{E}}[\lambda_k] = \bar{Y}$, $\widehat{\text{Var}}(\lambda_k) = \widehat{\text{Var}}(Y) - \bar{Y}$.

Linear Regression for $f(\text{gc})$ (Lec 4)

OLS: $\hat{\beta}_1 = \widehat{\text{Cov}}(\text{gc}, Y) / \widehat{\text{Var}}(\text{gc})$; $R^2 = 1 - \text{RSS}/\text{TSS}$.

Residuals vs. gc should be flat; fan shape $\Rightarrow$ heteroscedasticity. CIs wrong under misspecification. Global polynomials sensitive to outliers.

Spline Regression (Lec 4)

Progression: piecewise constant $\rightarrow$ discontinuous linear $\rightarrow$ broken stick (continuous) $\rightarrow$ cubic spline (continuous 1st & 2nd deriv.).

Cubic spline basis ($(u)_+ = \max(u, 0)$):

$$\hat{y} = b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \sum_j c_j (x - t_j)_+^3$$

B-spline bs(x, knots, degree=3): same fit, local support.

Broken stick (degree-1, knot t): slope is β_1 for $x < t$, and $\beta_1 + \beta_2$ for $x > t$. Per 0.01 increase: divide slope by 100.

Residual slope interpretation: positive slope in resid vs x for $x > t \Rightarrow$ true $f(x)$ rises faster than $\hat{f}(x)$ there (model underestimates).

Heteroscedasticity: $\text{SD ratio} = \text{SD}_{\text{hi}} / \text{SD}_{\text{lo}} \gg 1 \Rightarrow$ variance grows with coverage.

Poisson GLM & Quasi-Poisson (Lab 5)

Poisson GLM: $\log \mathbb{E}[Y_k] = \beta_0 + \beta_1 gc_k$ (log link, counts).

Assumes $\text{Var}(Y) = \mathbb{E}[Y]$ — still violated ($\phi \approx 14$).

Quasi-Poisson: same mean, $\text{Var}(Y) = \phi \mathbb{E}[Y]$, ϕ estimated from data. More honest uncertainty.

Comparing models: R^2 for lm; McFadden pseudo- R^2 for GLM; $\tilde{R}^2 = 1 - \frac{\log L(\hat{\text{model}})}{\log L(\text{null})}$.

Spearman r is rank-based — robust to outliers, but misleading when there is no real signal (centromere paradox: Spearman ≈ 0.95 despite $R^2 \approx 0$).

Bin size tradeoff: larger bins $\downarrow$ Poisson noise, $\uparrow$ GC signal blurring. Statistical elbow ~ 5 kb. Biological constraint: genes 10–100 kb.

Copy Number Estimation (Labs 6–8, Yuval's notes)

Model: $Y_k \sim \text{Pois}(\lambda_k)$,

$$\lambda_k = a_k \cdot f(\text{gc}_k) \cdot \epsilon_k$$

$a_k \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots\}$ = copy number. $f(\text{gc}_k)$ = expected coverage at copy number 1. ϵ_k = residual technical noise, $\mathbb{E}[\epsilon_k] = 1$.

Identifiability: $\text{med}_k\{a_k\} = 1$, $\mathbb{E}_k[f(\text{gc}_k)] = 1$.

Single-sample plug-in estimator $\hat{a}_k^{(1)}$:

$$\hat{a}_k^{(1)} = \frac{Y_k}{\hat{f}(\text{gc}_k)} \cdot \hat{M}, \quad \hat{M} = \frac{1}{\text{med}_k\left\{\frac{Y_k}{\hat{f}(\text{gc}_k)}\right\}}$$

Stabilized estimator (reduces variance, increases bias for $a_k \neq 1$):

$$\hat{a}_k^{(1*)} = \frac{Y_k + c}{\hat{f}(\text{gc}_k) + c}$$

Var decomp: $\text{Var}(\hat{a}_k) \approx \frac{1}{\hat{f}(\text{gc}_k)} + \frac{\sigma^2}{\hat{f}(\text{gc}_k)^2}$ (16% Poisson, 84% ϵ noise).

Two-sample model: shared γ_k , sample-specific δ_k^T, δ_k^N :

$$Y_k^T = a_k \cdot f^T(\text{gc}_k) \cdot \gamma_k \cdot \delta_k^T + \eta^T$$

$$Y_k^N = f^N(\text{gc}_k) \cdot \gamma_k \cdot \delta_k^N + \eta^N$$

Two-sample estimator:

$$\hat{a}_k^{(2)} = \frac{Y_k^T / \hat{f}^T(\text{gc}_k) + c}{Y_k^N / \hat{f}^N(\text{gc}_k) + c} \approx a_k \cdot \frac{\delta_k^T}{\delta_k^N}$$

Shared noise γ_k cancels. Residual noise is ratio of sample-specific terms only — better than single-sample if γ_k dominates.

Fit separate GC spline per sample. Ratio also cancels germline CNVs.

Noise: Additive $\lambda_k = a_k f + \epsilon$ vs. Multiplicative $\lambda_k = a_k f \cdot \eta$ — PCR $\Rightarrow$ multiplicative more plausible.

Poisson irreducible noise floor:

$$\mathbb{E}[(Y_k - f(\text{gc}_k))^2] = \mathbb{E}[(Y_k - \lambda_k)^2] \geq \mathbb{E}[\lambda_k] = \bar{\lambda}$$

Training MSE $< \bar{\lambda} \Rightarrow$ overfitting (memorizing noise).

Key diagnostic: corrected ratio vs position — flat at 1 under H_0 .

Aggregative model (Lab 7): fit at 1 kb, aggregate predictions to 10 kb bins.

Hypothesis Testing & Multiple Testing (Lab 9)

Statistic: $S_k = \log_2(\hat{a}_k^{2\text{-sample}})$. Under H_0 : $S_k \approx 0$.

Negative controls: locally stable bins (window median $\leq \log_2 1.10$, SD $\leq \log_2 1.15$), filtered on the *ratio* so germline CNVs pass correctly.

Empirical p-value: $p_k = |\{j : |S_j| \geq |S_k|\}| / |\text{ctrl}|$. No distributional assumption.

Calibration: P-P plot of held-out CN=1 bin p-values should track the uniform diagonal.

Bonferroni (FWER): $\alpha' = \alpha/m$. Too conservative ($m \sim 10^5$).

Benjamini–Hochberg (FDR): reject $H_{(k)}$ for $k \leq k^* = \max\{k : p_{(k)} \leq k\alpha/m\}$.

Controls $\mathbb{E}[\text{FP}/\text{sig}] \leq \alpha$. Preferred: tolerates a small known FP fraction for much greater power.

R Code Reference

Coverage counting (fast):

```
tabulate(positions) # counts per base, 0(n)
```

VMR & correlation:

```
vmr <- var(bins) / mean(bins)
cor(gc, cov, method="spearman") # rank-based
```

Piecewise discontinuous (4 params, X-row for GC= 0.3: (1, 0.3, 0, 0)):

```
lm(Y ~ I(GC<0.4) + I(GC*(GC<0.4)) +
    I(GC>=0.4) + I(GC*(GC>=0.4)))
```

Broken stick (continuous, 3 params, hinge at knot t):

```
lm(Y ~ GC + I(pmax(GC - t, 0)))
```

Cubic B-spline / GLM:

```
knots <- quantile(gc, c(.25,.5,.75))
lm(reads ~ bs(gc, knots=knots, degree=3))
glm(reads ~ bs(gc,...), family=quasipoisson())
```

GC correction & CN estimator:

```
fhat <- predict(fit, data.frame(gc=df$gc))
raw <- df$reads / fhat
a_hat <- raw / median(raw) # single-sample
cn_2s <- a_hat_tumor / a_hat_norm # two-sample
```

Multiple testing:

```
p.adjust(pvals, method="BH") # FDR
p.adjust(pvals, method="bonferroni")
```